Notes: La Porta, López-de-Silanes, Shleifer, and Vishny (JF, 1997)

Legal Determinants of External Finance

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1 Big picture

- **Question.** Why do some countries have much larger and broader capital markets than others? The paper asks whether cross-country differences in external finance can be explained by legal protection of investors and the quality of legal enforcement.
- **Setting.** The sample contains 49 countries. The paper compares common law countries with French, German, and Scandinavian civil law countries, and links legal origin and investor-protection measures to equity and debt market development.
- **Core idea.** Outside investors provide funds only when they expect to receive returns rather than be expropriated by insiders. Strong shareholder and creditor rights, backed by law enforcement, make outside finance more attractive and expand capital markets.
- **Main equity result.** Common law countries have larger equity markets, more listed firms per capita, and more IPOs per capita than civil law countries, especially French civil law countries.
- **Main debt result.** Countries with stronger creditor protection and better law enforcement tend to have broader debt markets, although creditor-rights variables are less powerful than shareholder-rights variables in explaining cross-country differences.
- **Legal origin result.** French civil law countries generally have the weakest investor protections and least developed capital markets. Common law countries have the strongest investor protections and broadest markets.
- **Law enforcement result.** Rule of law is strongly associated with both equity and debt market development. Rules matter, but enforcement matters too.
- **Economic takeaway.** Financial development is not only about savings, technology, or macroeconomic growth. The institutional environment that protects outside investors is a first-order determinant of whether firms can raise external finance.

2 Incremental contribution of the paper

- **From law and finance measurement to market outcomes.** The authors' earlier law-and-finance work measured investor-protection rules across countries. This paper connects those legal measures directly to capital-market size, breadth, and access to external finance.
- **A cross-country institutional explanation for financial development.** The paper proposes that legal origin and investor protection are deep determinants of financial development, not merely background country characteristics.
- **Equity market breadth, not just valuation.** The paper goes beyond stock-market capitalization by using listed firms per capita and IPOs per capita. This distinction matters because total market capitalization can be dominated by a few very large firms.
- **Debt as well as equity.** The analysis covers both external equity and private debt/bond finance. This gives a broader view of external finance than a pure stock-market study.
- **Rules versus enforcement.** The paper separates the content of legal rules from the quality of enforcement using measures such as antidirector rights, creditor rights, and rule of law.

- **Legal origin as a comparative framework.** The paper shows systematic differences across English common law, French civil law, German civil law, and Scandinavian civil law countries.
- **Large-firm versus aggregate finance.** By comparing aggregate measures with WorldScope firm-level medians, the paper shows that large firms may obtain finance even in countries where aggregate markets are narrow. This suggests that the main constraint may bind especially for smaller or less connected firms.

3 Data and measurement

3.1 Country sample and legal origin

- The main cross-country sample contains 49 countries.
- Countries are grouped by legal origin: English common law, French civil law, German civil law, and Scandinavian civil law.
- Legal origin is used as a summary proxy for broad legal traditions that affect investor protection and legal institutions.

Interpretation: Legal origin is not a randomly assigned treatment, but it is historically predetermined relative to modern capital market development. It therefore provides a useful organizing variable for institutional comparison.

3.2 Measures of external equity finance

The paper uses three main aggregate equity-market measures:

- **External cap/GNP.** Stock market capitalization held by minority outside investors divided by GNP. This adjusts market capitalization for insider ownership.
- **Domestic firms/Pop.** Number of domestic listed firms per million people.
- **IPOs/Pop.** Number of initial public offerings per million people during 1995:7–1996:6.

Conceptually,

$$\text{External cap/GNP}_c = \frac{\text{Stock market capitalization}_c \times (1 - \text{insider ownership}_c)}{\text{GNP}_c}.$$

Interpretation: The adjustment for insider holdings is important because high market capitalization does not necessarily mean large outside finance if insiders retain most shares.

3.3 Measures of debt finance

- **Debt/GNP.** Bank debt of the private sector plus outstanding nonfinancial bonds divided by GNP.
- At the firm level, the paper also studies the median ratios of debt to sales and debt to cash flow using WorldScope firms.

Interpretation: The aggregate debt measure captures the general ability of the private sector to borrow, not just the borrowing of listed firms.

3.4 Investor protection and legal enforcement

- **Rule of law.** A 0–10 index of the strength of the law-and-order tradition.
- **Antidirector rights.** A 0–5 index aggregating shareholder rights, such as proxy voting by mail, cumulative voting, oppressed-minority mechanisms, and the ability to call extraordinary meetings.
- **One-share-one-vote.** A dummy for whether ordinary shares legally carry one vote per share.
- **Creditor rights.** A 0–4 index aggregating creditor rights in liquidation and reorganization, including priority and restrictions on debtor control.

Interpretation: The central empirical hypothesis is that higher values of these protection/enforcement variables should be associated with broader and more valuable capital markets.

3.5 Controls

- **GDP growth:** average annual per capita GDP growth from 1970 to 1993.
- **Log GNP:** log gross national product in 1994.
- GDP per capita is generally not included because it is highly correlated with rule of law.

4 Models and empirical specifications

4.1 Legal-origin comparisons

The paper first compares mean external-finance outcomes across legal origins:

$$\bar{Y}_g = \frac{1}{N_g} \sum_{c \in g} Y_c,$$

where g is a legal-origin group. The main comparisons are common law versus civil law, and English versus French, German, and Scandinavian legal origins.

Interpretation: These comparisons show raw institutional patterns before controlling for growth, country size, and rule of law.

4.2 Investor-rights sorts

The paper sorts countries into bottom quartile, middle two quartiles, and top quartile by investor-protection measures:

$$\bar{Y}_{\text{top}} - \bar{Y}_{\text{bottom}}.$$

Interpretation: These simple sorts ask whether countries with stronger investor protections have larger capital markets without imposing a regression structure.

4.3 Cross-country OLS regressions

The main regression framework can be written as

$$Y_c = \alpha + \beta_1 \text{GDPGrowth}_c + \beta_2 \log(\text{GNP}_c) + \beta_3 \text{RuleOfLaw}_c + \Gamma' \text{Origin}_c + \Delta' \text{Rights}_c + \varepsilon_c.$$

- Y_c is one of the external-finance outcomes.
- Origin_c includes legal-origin dummies, with English common law as the omitted category.
- Rights_c includes antidirector rights, one-share-one-vote, or creditor rights depending on the specification.

Interpretation: The coefficients on origin and rights ask whether legal institutions predict capital market development after controlling for growth, country scale, and legal enforcement.

4.4 Equity-market regressions

The equity-market outcomes are:

$$Y_c \in \{\text{External cap/GNP},\ \text{Domestic firms/Pop},\ \text{IPOs/Pop}\}.$$

The key expectation is that stronger shareholder protection and common law origin should increase each equity-market outcome.

4.5 Debt-market regressions

The debt-market outcome is

$$Y_c = \text{Debt/GNP}_c.$$

The key expectation is that stronger creditor rights and rule of law should increase the availability of private debt finance.

4.6 Firm-level descriptive analysis

Using WorldScope, the paper studies country medians for:

$$\frac{\text{Market cap}}{\text{Sales}},\quad \frac{\text{Market cap}}{\text{Cash flow}},\quad \frac{\text{Debt}}{\text{Sales}},\quad \frac{\text{Debt}}{\text{Cash flow}}.$$

Interpretation: These firm-level ratios help assess whether the aggregate results are visible among the largest listed firms. The paper finds the equity patterns are similar but weaker, while large-firm debt ratios vary less by legal origin.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies cross-country associations between legal institutions and external finance.

- Legal origin is historically predetermined and helps organize institutional differences, but it is not randomly assigned.
- The paper is strongest as evidence that legal investor protection is systematically related to financial development.

Interpretation: The empirical design is institutional and comparative, not a natural experiment.

5.2 Main threats to interpretation

- **Omitted institutional factors.** Legal origin may proxy for trust, state capacity, political structure, colonial history, or broader institutional development.
- **Endogenous enforcement.** Richer and more financially developed countries may invest more in legal enforcement, making rule of law partly endogenous.
- **Measurement error.** Legal indices are coarse. The paper's own variables may not capture all relevant rights and protections.
- **Large-firm bias.** WorldScope covers primarily large listed firms, which may have access to external finance even in countries where smaller firms do not.

5.3 Why the evidence is still informative

- Multiple outcomes tell the same story: capitalization, listed firms, IPOs, and debt are all related to legal environment.
- The patterns are consistent with the mechanism that outside investors supply more capital when legally protected.
- The results align with the authors' earlier findings on investor protection and ownership concentration.

6 Tables and figures

6.1 Table I: Description of the Variables

Read:

- Table I defines all outcome variables, controls, and legal-institution measures.
- The paper uses both aggregate market measures and WorldScope firm-level measures.
- External cap/GNP adjusts stock-market capitalization for estimated insider ownership, which makes the measure closer to true outside equity finance.
- The legal variables separate legal origin, rule of law, shareholder rights, and creditor rights.

Economic message: Table I is the measurement foundation of the paper. It shows that the paper is not simply comparing raw market size; it builds variables designed to capture the actual external finance available to outside investors and firms.

Table I

Description of the Variables

Origin	Identifies the legal origin of the Company Law or Commercial Code of each country. Source: Reynolds and Flores (1989) and La Porta *et al*. (1996).
External cap/ GNP	The ratio of the stock market capitalization held by minorities to gross national product for 1994. The stock market capitalization held by minorities is computed as the product of the aggregate stock market capitalization and the average percentage of common shares not owned by the top three shareholders in the ten largest non-financial, privately-owned domestic firms in a given country. A firm is considered privately owned if the State is not a known shareholder in it. Source: *Moodys International*, *CIFAR*, *EXTEL*, *WorldScope*, *20-Fs*, *Price-Waterhouse*, and various country sources.
Domestic firms/ Pop	Ratio of the number of domestic firms listed in a given country to its population (in millions) in 1994. Source: *Emerging Market Factbook* and *World Development Report 1996*.
IPOs/Pop	Ratio of the number of initial public offerings of equity in a given country to its population (in millions) for the period 1995:7–1996:6. Source: *Securities Data Corporation*, *AsiaMoney*, *LatinFinance*, *GT Guide to World Equity Markets*, and *World Development Report 1996*.
Debt/GNP	Ratio of the sum of bank debt of the private sector and outstanding non-financial bonds to GNP in 1994, or last available. Source: *International Financial Statistics*, *World Bondmarket Factbook*.
GDP growth	Average annual percent growth of per capita gross domestic product for the period 1970–1993. Source: *World Development Report 1995*.
Log GNP	Logarithm of the Gross National Product in 1994. Source: *World Development Report 1996*.
Rule of law	Assessment of the law and order tradition in the country. Average of the months of April and October of the monthly index between 1982 and 1995. Scale from 0 to 10, with lower scores for less tradition for law and order. Source: *International Country Risk Guide*.
Antidirector rights	An index aggregating shareholder rights. The index is formed by adding 1 when: (1) the country allows shareholders to mail their proxy vote; (2) shareholders are not required to deposit their shares prior to the General Shareholders' Meeting; (3) cumulative voting is allowed; (4) an oppressed minorities mechanism is in place; or (5) when the minimum percentage of share capital that entitles a shareholder to call for an Extraordinary Shareholders' Meeting is less than or equal to 10% (the sample median). The index ranges from 0 to 5. Source: Company Law or Commercial Code and La Porta *et al*. (1996).
One-share = one-vote	Equals one if the Company Law or Commercial Code of the country requires that ordinary shares carry one vote per share, and 0 otherwise. Equivalently, this variable equals one when the law prohibits the existence of both multiple-voting and non-voting ordinary shares and does not allow firms to set a maximum number of votes per shareholder irrespective of the number of shares she owns, and 0 otherwise. Source: Company Law or Commercial Code and La Porta *et al*. (1996).

(a) Table I, part 1: variable definitions.

Table I—Continued

Creditor rights	An index aggregating creditor rights. The index is formed by adding 1 when: (1) the country imposes restrictions, such as creditors' consent or minimum dividends, to file for reorganization; (2) secured creditors are able to gain possession of their security once the reorganization petition has been approved (no automatic stay); (3) the debtor does not retain the administration of its property pending the resolution of the reorganization; (4) secured creditors are ranked first in the distribution of the proceeds that result from the disposition of the assets of a bankrupt firm. The index ranges from 0 to 4. Source: Company Law or Bankruptcy Laws and La Porta *et al*. (1996).
Market cap/ sales	The median ratio of the stock market capitalization held by minorities to sales in 1994 for all nonfinancial firms in a given country on the *WorldScope* database. Firm's *j* stock market capitalization held by minorities is computed as the product of the stock market capitalization of firm *j* and the average percentage of common shares not owned by the top three shareholders in the ten largest nonfinancial, privately-owned domestic firms in a given country. A firm is considered privately owned if the State is not a known shareholder in it. Source: *WorldScope*.
Market cap/ cash-flow	The median ratio of the stock market capitalization held by minorities to cash flow in 1994 for all nonfinancial firms in a given country on the *WorldScope* database. Firm's *j* stock market capitalization held by minorities is computed as the product of the stock market capitalization of firm *j* and the average percentage of common shares not owned by the top three shareholders in the ten largest nonfinancial, privately-owned domestic firms in a given country. A firm is considered privately owned if the State is not a known shareholder in it. Source: *WorldScope*.
Debt/sales	Median of the total-debt-to-sales ratio in 1994 for all firms in a given country on the *WorldScope* database. Source: *WorldScope*.
Debt/cash flow	Median of the total-debt-to-cash-flow ratio for all firms in a given country on the *WorldScope* database. Source: *WorldScope*.

(b) Table I, part 2: continued definitions.

6.2 Table II: External Capital Markets

Read:

- Table II reports country-level external finance measures by legal origin.
- English-origin countries have much larger equity markets than French-origin countries: the English-origin average external cap/GNP is 0.60 versus 0.21 for French-origin countries.
- English-origin countries also have more listed firms per capita and more IPOs per capita.
- French civil law countries have the weakest antidirector and creditor-rights scores among the major legal-origin groups.
- Debt/GNP is highest in German-origin countries, which is consistent with the view that German-origin countries rely more heavily on bank-centered finance.

Economic message: The raw data already reveal the central institutional pattern: common law countries have broader equity finance, while French civil law countries have weaker investor protection and less developed capital markets.

Table II

External Capital Markets

This table classifies countries by legal origin. Definitions for each of the variables can be found in Table I. Panel B reports tests of means for the different legal origins.

Country	External Cap/GNP	Domestic Firms/Pop	IPOs/Pop	Debt/GNP	GDP growth	Log GNP	Rule of Law	Antidirector Rights	One-Share = One-Vote	Creditor Rights
Panel A: Means										
Australia	0.49	63.55	–	0.76	3.06	12.64	10.00	4	0	1
Canada	0.39	40.86	4.93	0.72	3.36	13.26	10.00	4	0	1
Hong Kong	1.18	88.16	5.16	–	7.57	11.56	8.22	4	1	4
India	0.31	7.79	1.24	0.29	4.34	12.50	4.17	2	0	4
Ireland	0.27	20.00	0.75	0.38	4.25	10.73	7.80	3	0	1
Israel	0.25	127.60	1.80	0.66	4.39	11.19	4.82	3	0	4
Kenya	–	2.24	–	–	4.79	8.83	5.42	3	0	4
Malaysia	1.48	25.15	2.89	0.84	6.90	11.00	6.78	3	1	4
New Zealand	0.28	69.00	0.66	0.90	1.67	10.69	10.00	4	0	3
Nigeria	0.27	1.68	–	–	3.43	10.36	2.73	3	0	4
Pakistan	0.18	5.88	–	0.27	5.50	10.88	3.03	4	1	4
Singapore	1.18	80.00	5.67	0.60	1.68	11.68	8.57	3	1	3
South Africa	1.45	16.00	0.05	0.93	7.48	10.92	4.42	4	0	4
Sri Lanka	0.11	11.94	0.11	0.25	4.04	9.28	1.90	2	0	3
Thailand	0.56	6.70	0.56	0.93	7.70	11.72	6.25	3	0	3
UK	1.00	35.68	2.01	1.13	2.27	13.86	8.57	4	0	4
US	0.58	30.11	3.11	0.81	2.74	15.67	10.00	5	0	1
Zimbabwe	0.18	5.81	–	–	2.17	8.63	3.68	3	0	4
English origin avg	**0.60**	**35.45**	**2.23**	**0.68**	**4.30**	**11.41**	**6.46**	**3.39**	**0.22**	**3.11**
Argentina	0.07	4.58	0.20	0.19	1.40	12.40	5.35	4	0	1
Belgium	0.17	15.50	0.30	0.38	2.46	12.29	10.00	0	0	2
Brazil	0.18	3.48	0.00	0.39	3.95	13.03	6.32	3	1	1
Chile	0.80	19.92	0.35	0.63	3.35	10.69	7.02	3	1	2
Colombia	0.14	3.13	0.05	0.19	4.38	10.82	2.08	1	0	0
Ecuador	–	13.18	0.09	–	4.55	9.49	6.67	2	0	4
Egypt	0.08	3.48	–	–	6.13	10.53	4.17	2	0	4
France	0.23	8.05	0.17	0.96	2.54	14.07	8.98	2	0	0
Greece	0.07	21.60	0.30	0.23	2.46	11.25	6.18	1	1	1
Indonesia	0.15	1.15	0.10	0.42	6.38	11.84	3.98	2	0	4
Italy	0.08	3.91	0.31	0.55	2.82	13.94	8.33	0	0	2
Jordan	–	23.75	–	0.70	1.20	8.49	4.35	1	0	–
Mexico	0.22	2.28	0.03	0.47	3.07	12.69	5.35	0	0	0
Netherlands	0.52	21.13	0.66	1.08	2.55	12.68	10.00	2	0	2
Peru	0.40	9.47	0.13	0.27	2.82	10.92	2.50	2	1	0
Philippines	0.10	2.90	0.27	0.10	0.30	10.44	2.73	4	0	0
Portugal	0.08	19.50	0.50	0.64	3.52	11.41	8.68	2	0	1
Spain	0.17	9.71	0.07	0.75	3.27	13.19	7.80	2	0	2
Turkey	0.18	2.93	0.05	0.15	5.05	12.08	5.18	2	0	2
Uruguay	–	7.00	0.00	0.26	1.96	9.40	5.00	1	1	2
Venezuela	0.08	4.28	0.00	0.10	2.65	10.99	6.37	1	0	–
French origin avg	**0.21**	**10.00**	**0.19**	**0.45**	**3.18**	**11.55**	**6.05**	**1.76**	**0.24**	**1.58**
Austria	0.06	13.87	0.25	0.79	2.74	12.13	10.00	2	0	3
Germany	0.13	5.14	0.08	1.12	2.60	14.46	9.23	1	0	3
Japan	0.62	17.78	0.26	1.22	4.13	15.18	8.98	3	1	2
South Korea	0.44	15.88	0.02	0.74	9.52	12.73	5.35	2	1	3
Switzerland	0.62	33.85	–	–	1.18	12.44	10.00	1	0	1
Taiwan	0.88	14.22	0.00	–	11.56	12.34	8.52	3	0	2
German origin avg	**0.46**	**16.79**	**0.12**	**0.97**	**5.29**	**13.21**	**8.68**	**2.00**	**0.33**	**2.33**
Denmark	0.21	50.40	1.80	0.34	2.09	11.84	10.00	3	0	3
Finland	0.25	13.00	0.60	0.75	2.40	11.49	10.00	2	0	1
Norway	0.22	33.00	4.50	0.64	3.43	11.62	10.00	3	0	2
Sweden	0.51	12.66	1.66	0.55	1.79	12.28	10.00	2	0	2
Scandinavian origin avg	**0.30**	**27.26**	**2.14**	**0.57**	**2.42**	**11.80**	**10.00**	**2.50**	**0.00**	**2.00**
Sample average	**0.40**	**21.59**	**1.02**	**0.59**	**3.79**	**11.72**	**6.85**	**2.44**	**0.22**	**2.30**

(a) Table II, part 1: country means by legal origin.

Table II—Continued

Country	External Cap/GNP	Domestic Firms/Pop	IPOs/Pop	Debt/GNP	GDP growth	Log GNP	Rule of Law	Antidirector Rights	One-Share = One-Vote	Creditor Rights
Panel B: Tests of Means (*t*-statistics)										
Common vs civil law	3.12	3.16	3.97	1.33	1.23	−1.06	−0.77	5.24	−0.03	3.61
English vs French origin	3.29	3.16	4.50	2.29	1.97	−0.28	0.51	5.13	−0.11	3.61
English vs German origin	0.68	1.24	2.34	−1.88	−0.78	−2.31	−1.82	3.66	−0.52	1.43
English vs Scand. origin	1.25	0.44	0.08	0.71	1.81	−0.44	−15.57	2.14	2.20	1.71
French vs German origin	−2.38	−1.85	0.78	−3.39	−1.96	−2.48	−2.55	−0.47	−0.45	−1.29
French vs Scand. origin	−0.91	−3.31	−5.45	0.82	0.97	−0.33	−20.80	−1.25	2.50	−0.60
German vs Scand. origin	0.94	−1.21	−2.76	2.71	1.32	2.11	−11.29	−0.98	1.58	0.63

(b) Table II, part 2: tests of legal-origin means.

6.3 Table III: Investor Rights and External Finance

Read:

- Table III sorts countries by antidirector rights, one-share-one-vote, creditor rights, rule of law, GDP growth, and log GNP.
- Countries in the top quartile of antidirector rights have higher external cap/GNP, more domestic firms per capita, and more IPOs per capita than countries in the bottom quartile.
- Rule of law is especially important for market breadth: countries in the top rule-of-law quartile have many more domestic firms and IPOs per capita.
- Debt/GNP is much higher in countries with stronger rule of law.

Economic message: The sorting evidence links investor protection directly to the ability of firms to raise outside finance. Stronger legal protection is associated not only with higher valuations but also with more firms accessing public markets.

6.4 Table IV: External Market Capitalization of Equity/GNP Regressions

Read:

- Table IV estimates cross-country OLS regressions for external cap/GNP.
- GDP growth is positively related to external market capitalization.
- Rule of law is positive across specifications.
- Civil law origin dummies are negative relative to English common law.
- Antidirector rights and one-share-one-vote are positive; when included together with origin, they explain part of the legal-origin differences but not all of them.

Economic message: The valuation of outside equity is higher when legal rules and enforcement protect outside shareholders. Legal origin still matters after controls, implying that measured shareholder-rights variables do not capture the full institutional difference.

6.5 Table V: Domestic Firms/Population Regressions

Read:

- Table V uses the number of domestic listed firms per million people as the dependent variable.
- Rule of law is strongly positive and statistically significant.
- Log GNP is negative, implying that larger economies do not mechanically have more listed firms per capita.
- French, German, and Scandinavian legal-origin dummies are negative relative to English common law.
- Antidirector rights are positive when included alone, but lose explanatory power when legal-origin dummies are included.

Table III

Investor Rights and External Finance

This table classifies countries according to their ranking in: (a) Antidirector Rights; (b) One-Share = One-Vote; (c) Creditor Rights; (4) Rule of Law; (5) GDP Growth; and (5) Log GNP. For each panel, the table shows the average value of different external finance measures for the bottom quartile, the middle two quartiles, and the top quartile. The last row of each panel shows the *t*-statistic for a test of means between the bottom and the top quartiles.

	External Cap/GNP	Domestic Firms/Pop	IPOs/Pop	Debt/GNP
Means by antidirector rights				
Bottom 25%	0.19	12.05	0.14	0.44
Mid 50%	0.39	20.03	0.97	0.63
Top 25%	0.58	35.68	2.05	0.63
Test of means (*t*-statistic)				
Bottom 25% vs. Top 25%	−2.50	−2.35	−2.55	−1.22
Means by one-share = one-vote				
Not One Vote	0.32	20.10	0.87	0.59
One Vote	0.65	26.76	1.48	0.56
Test of means (*t*-statistic)				
One Vote vs Not One Vote	−2.61	−0.76	−1.08	0.29
Means by creditor rights				
Bottom 25%	0.27	18.43	0.85	0.49
Mid 50%	0.40	18.25	0.62	0.66
Top 25%	0.59	31.30	2.37	0.65
Test of means (*t*-statistic)				
Bottom 25% vs. Top 25%	−2.09	−1.11	−1.95	−1.15
Means by rule of law				
Bottom 25%	0.28	8.51	0.28	0.34
Mid 50%	0.47	22.36	0.89	0.63
Top 25%	0.36	33.08	1.85	0.70
Test of means (*t*-statistic)				
Bottom 25% vs. Top 25%	−0.73	−4.11	−2.30	−3.84
Means by GDP growth				
Bottom 25%	0.42	22.83	0.74	0.54
Mid 50%	0.28	15.90	0.86	0.60
Top 25%	0.62	30.43	1.64	0.62
Test of means (*t*-statistic)				
Bottom 25% vs. Top 25%	−1.05	−0.61	−1.20	−0.56
Means by log GNP				
Bottom 25%	0.25	15.36	0.27	0.43
Mid 50%	0.46	26.12	1.33	0.50
Top 25%	0.39	19.82	0.98	0.82
Test of means (*t*-statistic)				
Bottom 25% vs. Top 25%	−1.31	−0.63	−1.24	−3.26

(a) Table III: investor rights and external finance.

Table IV

External Market Capitalization of Equity/GNP Regressions

Ordinary least squares regressions of the cross-section of 49 countries around the world. The dependent variable is "External Cap." The independent variables are (1) GDP Growth; (2) Log GNP; (3) Rule of law; (4) French origin; (5) German origin; (6) Scandinavian origin; (7) Antidirector Rights; (8) One-share = One-Vote. Standard errors are shown in parentheses.

Independent Variables	Dependent Variable: External Cap/GNP				
GDP growth	0.0617^{b}	0.0544^{b}	0.0584^{b}	0.0562^{b}	0.0441^{b}
	(0.0232)	(0.0201)	(0.0238)	(0.0242)	(0.0209)
Log GNP	−0.0129	−0.0168	0.0038	−0.0053	0.0091
	(0.0333)	(0.0334)	(0.0386)	(0.0382)	(0.0324)
Rule of law	0.0378^{c}	0.0455^{b}	0.0417	0.0424^{b}	0.0437^{c}
	(0.0206)	(0.0203)	(0.0250)	(0.0243)	(0.0231)
French origin			-0.3225^{a}	-0.2142^{c}	-0.3341^{a}
			(0.1131)	(0.1194)	(0.1084)
German origin			-0.2962^{c}	−0.1849	-0.3230^{b}
			(0.1497)	(0.1599)	(0.1438)
Scandinavian origin			-0.3391^{b}	-0.2816^{c}	-0.3056^{b}
			(0.1373)	(0.1479)	(0.1218)
Antidirector rights	0.1171^{a}			0.0675^{c}	
	(0.0353)			(0.0354)	
One-share = one-vote		0.2745^{b}			0.2890^{b}
		(0.1235)			(0.1111)
Intercept	−0.2437	0.0100^{b}	0.0336	−0.0860	−0.0475
	(0.2880)	(0.3063)	(0.3677)	(0.3629)	(0.3066)
Observations	45	45	45	45	45
Adjusted R^2	0.2936	0.2347	0.2867	0.3016	0.3801

[a] Significant at 1%; [b] Significant at 5%; [c] Significant at 10%.

(b) Table IV: external capitalization/GNP regressions.

Economic message: The breadth of equity markets is much greater in legal environments that protect outside investors. The finding is about access to public markets, not just high prices for a few listed firms.

6.6 Table VI: Initial Public Offerings/Population Regressions

Read:

- Table VI studies IPOs per million people.
- Rule of law is positive and significant.
- Antidirector rights are positive and significant, and one-share-one-vote becomes significant in the full specification.
- French and German civil law countries have substantially fewer IPOs than common law countries.
- Scandinavian-origin countries are not consistently different from common law countries in the IPO regressions.

Economic message: IPO activity is one of the clearest signs of external equity access. The results suggest that strong shareholder protection supports new public equity issuance.

Table V

Domestic Firms/Population Regressions

Ordinary least squares regressions of the cross-section of 49 countries around the world. The dependent variable is "Domestic Firms/Pop." The independent variables are (1) GDP growth; (2) Log GNP; (3) Rule of law; (4) French origin; (5) German origin; (6) Scandinavian origin; (7) Antidirector rights; (8) One-share = one-vote. Standard errors are shown in parentheses.

Independent Variables	Dependent Variable: Domestic Firms/Pop				
GDP growth	1.0767 (1.4000)	1.3461 (1.3318)	1.0111 (1.2661)	0.8950 (1.2733)	0.5763 (0.9884)
Log GNP	−4.3181[b] (1.6588)	−4.0659[b] (1.7697)	−2.9126 (1.7698)	−3.3073[b] (1.8165)	−2.7979 (1.6816)
Rule of law	4.5093[a] (1.2579)	4.8584[a] (1.4023)	4.8422[a] (1.3616)	4.8577[a] (1.3377)	4.9582[a] (1.3356)
French origin			−21.9069[a] (7.4014)	−17.5313[b] (8.9183)	−22.5204[a] (7.2884)
German origin			−25.1485[a] (8.4882)	−20.5611[b] (9.7216)	−26.3007[a] (7.8639)
Scandinavian origin			−22.2680[b] (10.1744)	−19.9575[c] (10.0144)	−21.3009[b] (10.0541)
Antidirector rights	7.3034[a] (1.8052)			2.7304 (1.6591)	
One-share = one-vote		8.1382 (7.5228)			10.0675 (6.3165)
Intercept	19.3863 (15.4445)	29.0780[c] (17.0108)	33.0485 (20.6317)	28.6987 (21.4015)	30.6212 (20.2510)
Number of observations	49	49	49	49	49
Adjusted R^2	0.2198	0.1153	0.2197	0.2495	0.2681

[a] Significant at 1%; [b] Significant at 5%; [c] Significant at 10%.

(a) Table V: domestic firms/population regressions.

Table VI

Initial Public Offerings/Population Regressions

Ordinary least squares regressions of the cross-section of 49 countries around the world. The dependent variable is "IPOs/Pop." The independent variables are (1) GDP growth; (2) Log GNP; (3) Rule of law; (4) French origin; (5) German origin; (6) Scandinavian origin; (7) Antidirector rights; (8) One-share = one-vote. Standard errors are shown in parentheses.

Independent Variables	Dependent Variable: IPOs/Pop				
GDP growth	0.1222 (0.1281)	0.1320 (0.1193)	0.1937[b] (0.1012)	0.1916[c] (0.1037)	0.1633[b] (0.0744)
Log GNP	−0.1672 (0.1453)	−0.1225 (0.1692)	0.0662 (0.1086)	0.0452 (0.1129)	0.1255 (0.1002)
Rule of law	0.2549[a] (0.0889)	0.2943[a] (0.0926)	0.2122[b] (0.0842)	0.2108[b] (0.0830)	0.2127[a] (0.0731)
French origin			−1.5982[a] (0.3552)	−1.2949[a] (0.3696)	−1.6677[a] (0.3132)
German origin			−2.8118[a] (0.5698)	−2.5450[a] (0.5909)	−3.027[a] (0.5543)
Scandinavian origin			−0.3123 (0.8666)	−0.1421 (0.8486)	−0.1367 (0.8414)
Antidirector rights	0.5352[a] (0.1364)			0.1937[c] (0.0989)	
One-share = one-vote		0.6359 (0.5422)			1.0287[a] (0.3450)
Intercept	−0.5546 (1.3472)	−0.2720 (1.7534)	−0.9201 (1.3233)	−1.3071 (1.3204)	−1.7268 (1.2088)
Number of observations	41	41	41	41	41
Adjusted R^2	0.3082	0.1571	0.4907	0.4927	0.5643

[a] Significant at 1%; [b] Significant at 5%; [c] Significant at 10%.

(b) Table VI: IPOs/population regressions.

6.7 Table VII: Debt/GNP Regressions

Read:

- Table VII regresses aggregate private debt plus bonds over GNP on controls and legal variables.
- Rule of law is positive and highly significant across specifications.
- Creditor rights are positive when included without legal-origin dummies, but become weaker when origin dummies are added.
- French origin is associated with lower debt/GNP relative to common law in some specifications.
- German origin has a positive but statistically insignificant coefficient, consistent with large German-style banking systems.

Economic message: Debt finance depends strongly on enforcement. The creditor-rights index is less successful than shareholder-rights measures in explaining cross-country differences, perhaps because bank finance, state banks, and informal enforcement are harder to capture with simple legal indices.

6.8 Table VIII: External Funding at the Firm Level

Read:

- Table VIII uses WorldScope data for large listed firms in 38 countries.
- Large firms in common law countries have somewhat higher market-capitalization-to-sales ratios than large firms in several civil law groups.

- The cross-legal-origin equity differences are weaker at the firm level than in the aggregate data.
- Debt ratios of large firms are much more similar across legal origins than aggregate debt measures.
- Countries with high aggregate external cap/GNP also tend to have higher valuation ratios for their largest firms, but countries with low aggregate debt/GNP do not necessarily have low debt among their largest firms.

Economic message: The firm-level evidence suggests that the biggest listed firms can often obtain finance even in weak legal environments. The main aggregate effect of investor protection may operate through market breadth and access for a larger set of firms, not only the leverage or valuation of already-large firms.

Table VII

Debt/GNP Regressions

Ordinary least squares regressions of the cross-section of 49 countries around the world. The dependent variable is "Debt/GNP." The independent variables are (1) GDP growth; (2) Log GNP; (3) Rule of law; (4) French origin; (5) German origin; (6) Scandinavian origin; (7) Creditor rights. Standard errors are shown in parentheses.

Independent Variables	Dependent Variable: Debt/GNP		
GDP growth	0.0310[c]	0.0251[c]	0.0197
	(0.0171)	(0.0134)	(0.0152)
Log GNP	0.0667[b]	0.0370	0.0404
	(0.0252)	(0.0255)	(0.0250)
Rule of law	0.0615[a]	0.0698[a]	0.0694[a]
	(0.0132)	(0.0147)	(0.0148)
French origin		−0.1516[b]	−0.1163
		(0.0740)	(0.0825)
German origin		0.1080	0.1082
		(0.1010)	(0.0982)
Scandinavian origin		−0.2764[b]	−0.2618[b]
		(0.1037)	(0.1075)
Creditor rights	0.0518[c]		0.0270
	(0.0267)		(0.0298)
Intercept	−0.8621[a]	−0.3496	−0.4414
	(0.2579)	(0.2524)	(1.341)
Number of observations	39	39	39
Adjusted R^2	0.5522	0.5191	0.5984

[a] Significant at 1%; [b] Significant at 5%; [c] Significant at 10%.

(a) Table VII: debt/GNP regressions.

Table VIII

External Funding at the Firm Level

The sample of thirty-eight countries includes all the firms on the Worldscope database for 1996. The table shows median values for all the firms in each country. Panel A show the medians based on a classification by legal origin. The definition for each of the variables can be found in Table I. Panel B gives the tests of means for the different legal origins. Panel C shows mean of medians and *t*-tests for countries sorted by levels of "External Cap/GNP." Panel D shows mean of medians and *t*-tests for countries sorted by "Debt/GNP."

Country	Market Cap/Sales	Market Cap/ Cash-Flow	Debt/ Sales	Debt/Cash-Flow	WorldScope Firms/ Domestic Firms
Panel A: Median Values by Legal Origin					
Australia	0.75	6.15	0.19	1.42	0.12
Canada	0.76	4.66	0.30	2.07	0.26
Hong Kong	0.66	4.01	0.31	2.50	0.12
India	0.73	8.75	0.47	4.26	0.01
Ireland	0.75	3.51	0.16	0.74	0.29
Israel	0.34	3.79	0.17	1.41	0.03
Malaysia	1.46	6.82	0.24	1.45	0.23
New Zealand	0.38	4.26	0.23	2.74	0.11
Pakistan	0.50	4.18	0.33	2.34	0.05
Singapore	0.83	5.68	0.07	0.83	0.19
South Africa	0.40	3.23	0.29	2.06	0.22
Thailand	0.71	4.65	0.54	3.45	0.32
UK	0.64	5.77	0.11	1.06	0.51
US	0.67	6.70	0.18	1.86	0.28
Average English origin	**0.69**	**5.16**	**0.26**	**2.01**	**0.20**
Argentina	0.63	4.18	0.28	1.78	0.10
Belgium	0.16	2.28	0.25	2.52	0.39
Brazil	0.24	1.97	0.18	1.52	0.11
Chile	1.68	8.15	0.29	1.59	0.13
France	0.29	4.28	0.19	2.36	0.67
Greece	0.25	5.99	0.21	2.55	0.04
Indonesia	0.48	3.03	0.37	3.25	0.23
Italy	0.17	2.21	0.32	3.04	0.44
Mexico	0.47	4.06	0.66	1.54	0.29
Netherlands	0.27	3.93	0.11	1.33	0.42
Philippines	1.61	5.17	0.29	0.86	0.14
Portugal	0.19	2.48	0.33	3.73	0.17
Spain	0.27	3.28	0.25	2.33	0.15
Turkey	0.46	2.87	0.11	0.50	0.12
Average French origin	**0.51**	**3.85**	**0.27**	**2.06**	**0.24**
Austria	0.21	2.29	0.24	2.38	0.17
Germany	0.21	3.29	0.10	1.24	0.55
Japan	0.63	13.80	0.34	6.99	0.50
South Korea	0.29	—	0.58	—	0.09
Switzerland	0.26	3.06	0.30	3.14	0.36
Taiwan	2.21	14.94	0.26	2.16	0.20
Average German origin	**0.63**	**7.48**	**0.30**	**3.18**	**0.31**
Denmark	0.30	3.30	0.22	1.88	0.38
Finland	0.30	2.90	0.31	2.58	0.80
Norway	0.49	3.70	0.36	3.62	0.46
Sweden	0.40	3.10	0.21	1.59	0.82
Average Scandinavian origin	**0.37**	**3.25**	**0.28**	**2.42**	**0.61**
Sample average	**0.58**	**4.77**	**0.27**	**2.24**	**0.28**

(b) Table VIII, part 1: firm-level funding by legal origin.

7 Short summary

- The paper studies whether legal investor protection explains cross-country differences in external finance.
- The main sample covers 49 countries and classifies legal systems by origin: English, French, German, and Scandinavian.
- Equity-market outcomes include external cap/GNP, domestic listed firms per capita, and IPOs per capita.

Table VIII—Continued

Country	Market Cap/Sales	Market Cap/ Cash-Flow	Debt/ Sales	Debt/Cash-Flow	WorldScope Firms/ Domestic Firms
Panel B: Tests of Means (*t*-statistics)					
Common vs. civil law	1.04	0.64	−0.60	−0.87	
England vs. France	1.10	2.11	−0.36	−0.14	
England vs. Germany	0.20	−1.33	−0.71	−1.61	
England vs. Scandinavia	2.17	2.38	−0.32	−0.73	
France vs. Germany	−0.42	−2.04	−0.43	−1.59	
France vs. Scandinavia	0.54	0.69	−0.06	−0.68	
Germany vs. Scandinavia	0.64	1.32	0.30	0.64	
Panel C: Sorted by External Cap/GNP			Panel D: Sorted by Debt/GNP		
Means					
Bottom 25%	0.29	3.23	0.26	1.94	
Mid 50%	0.53	4.31	0.29	2.17	
Top 25%	0.97	7.28	0.24	2.52	
Test of means					
Bottom 25% vs. Top 25%	−3.03	−2.68	0.33	−0.80	

Table VIII, part 2: firm-level tests and aggregate-sorted comparisons.

- Debt-market outcomes include private bank debt plus corporate bonds over GNP.
- Key legal variables include rule of law, antidirector rights, one-share-one-vote, and creditor rights.
- Common law countries have stronger investor protections and broader capital markets, especially compared with French civil law countries.
- Rule of law strongly predicts both equity and debt market development.
- Shareholder-rights variables help explain equity market size and breadth.
- Creditor-rights variables are less robust in debt regressions, but enforcement remains powerful.
- Large firms can obtain debt even in countries with weak aggregate capital markets, implying that weak legal environments may restrict access especially for smaller or less privileged firms.

8 Conclusion

8.1 Core conclusion

The paper concludes that legal rules and legal enforcement are central determinants of financial development. Countries that protect outside investors have larger and broader capital markets, while countries with weaker investor protection have smaller and narrower external-finance markets.

8.2 Economic intuition

External finance requires outside investors to believe they will receive returns rather than be expropriated by insiders. Strong shareholder rights, creditor rights, and law enforcement reduce expropriation risk. As a result, outside investors are willing to provide more capital, entrepreneurs can raise funds on better terms, and more firms can access public markets.

8.3 Incremental contribution revisited

The paper turns legal origin and investor protection into an empirical explanation for cross-country financial development. It shows that law is not only a background institution; it is directly related to equity-market valuation, equity-market breadth, IPO activity, and private debt.

8.4 Implications

- **For financial development:** Building capital markets requires investor protection, not just macroeconomic growth.
- **For corporate finance:** The availability of outside finance depends on national legal institutions.
- **For comparative institutions:** Legal origin is a powerful predictor of financial-market structure.
- **For emerging markets:** Weak law enforcement and poor minority-investor protections can limit IPOs and public market breadth.

8.5 Limitations

- The design is cross-sectional and cannot fully establish causality.
- Legal origin may proxy for broader institutional, political, cultural, or trust-related variables.
- The legal indices are coarse and may miss important country-specific enforcement channels.
- The WorldScope sample covers mainly large firms, so it may understate financing constraints faced by smaller firms.
- The paper does not directly estimate real outcomes such as investment, productivity, or growth.

8.6 Takeaway

The main takeaway is that investor protection is a core input into financial development. Strong legal protection makes external finance credible; weak protection keeps ownership concentrated and capital markets narrow. The paper is foundational because it makes legal institutions a central variable in corporate finance and financial development.